COURSE GUIDE & SYLLABUS · GRADES 6–8 · AGES 11–14

Scratch (Gr 6–8) Intermediate: Data & Simulations

Lists, procedures, models and robust apps — think like a software engineer and a scientist.

The second middle-school Scratch course. Learners store and analyse data with lists (traversal, totals, averages, maximum and minimum), organise code with procedures, decomposition and refactoring, model the world with random simulations and agent-based models, and build robust data-driven apps with validation and menus. Projects: Survey Analyser, Procedural City, My Simulation and a capstone Data App. Every lesson links to the equivalent Python and includes a video, build-along, auto-checked Code Lab challenges, a worksheet and a teacher lesson plan.

What learners will be able to do

- Store data in lists and traverse them with an index variable
- Calculate totals, counts, averages, maximums and minimums
- Collect and analyse survey data responsibly
- Define procedures with parameters and explain parameter vs argument
- Decompose problems and refactor repeated code
- Simulate random events and explain the law of large numbers
- Build step-by-step and agent-based models and question their assumptions
- Design data-driven programs with parallel lists
- Validate input and build menus that handle every choice
- Read the Python equivalents of lists, functions and loops

How every lesson works

Watch	A short animated explainer video with captions (and an optional read-aloud).
Learn	Learn cards with the real coding blocks, and a step-by-step build-along.
Try it (Code Lab)	Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time.
Puzzles & quick checks	Robot puzzles and questions that build understanding, with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric.

Printables	A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher.

Scope and sequence

#	Lesson	Big idea	Learners build
1	**Lists and Traversal** Store data in a list and traverse it with a loop and an index variable to process every item.	Lists and traversal	Weekly temperatures
2	**Analysing Data: Average, Maximum, Minimum** Use traversal to calculate an average and find the maximum and minimum values in a list.	Data algorithms	Weather report
3	**Project: Survey Analyser** Plan, build and test an app that collects survey answers into a list and reports statistics.	Project: data app	Project: Survey Analyser
4	**Procedures with Inputs** Define custom blocks (procedures) that take an input, and call them with different arguments to reuse code.	Procedures and parameters	The polygon procedure
5	**Decomposition and Abstraction** Break a problem into smaller procedures, build a solution from them, and refactor repeated code into a procedure.	Decomposition and refactoring	A street of houses
6	**Project: Procedural City** Plan, build and test a program that draws a city skyline from procedures with parameters and randomness.	Project: procedures	Project: Procedural City
7	**Randomness and Probability** Use random numbers to simulate chance events many times, and compare experimental results with theoretical probability.	Random simulation	Coin-flip lab
8	**Models and Agents** Build a numeric model that changes over time, and an agent-based model where many clones follow simple rules.	Modelling	Rabbits and particles
9	**Project: My Simulation** Plan, build and test a simulation of a real-world process, run it as an experiment and report what you find.	Project: simulation	Project: My Simulation
10	**A Quiz Engine with Parallel Lists** Store questions and answers in parallel lists and write one loop that runs a quiz of any length.	Data-driven programs	Build the quiz engine
11	**Robust Apps: Validation and Menus** Make apps robust: re-ask until input is valid, handle every menu choice, and organise features into procedures.	Validation and program structure	Robust age checker and menu

12	**Capstone: Data App** Design, build, test and present a data-driven app that uses lists, procedures, validation and a menu or simulation.	Capstone project	Project: Data App

Modules

Module 1 · Lists and data
Traversal, accumulators, averages and extremes. Project: Survey Analyser.

Big question: How do programs turn raw data into insight?

Module 2 · Procedures and design
Parameters, decomposition, abstraction and refactoring. Project: Procedural City.

Big question: How do engineers keep big programs manageable?

Module 3 · Simulations
Probability, random walks, models and agents. Project: My Simulation.

Big question: How can code help us understand the real world?

Module 4 · Data-driven apps
Quiz engines with parallel lists, validation and menus, then a capstone Data App.

Big question: What makes an app robust and useful?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and the four projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Laptop or Chromebook with a modern browser and a keyboard
- The built-in Kitabash Code Lab (no account needed)
- Optional: Scratch (scratch.mit.edu)

Before you start

- Scratch (Gr 6–8) Beginner: Code Foundations, or confident use of loops, variables, if-else and the game loop

Safety and privacy

Surveys collect anonymous, non-sensitive answers only — never names with personal data. Sharing to scratch.mit.edu is optional and needs a parent-approved account for under-13s. Learners under 18 use the AI tutor only with guardian consent.

What comes next

After this course: **Scratch (Gr 6–8) Advanced**, which goes deeper into clones, custom blocks, lists and bigger games.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.